

TITANIC DATASET ANALYSIS USING EXPLORATORY DATA ANALYSIS (EDA) AND DATA VISUALIZATION

CodeAlpha Data Analytics Internship

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DECLARATION

I hereby declare that this project report entitled 'Titanic Dataset Analysis using Exploratory Data Analysis and Data Visualization' is my original work completed during the CodeAlpha Data Analytics Internship.

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ABSTRACT

This project analyzes the Titanic dataset using EDA and Data Visualization techniques to identify factors affecting passenger survival and derive meaningful insights.

INTRODUCTION

Data Analytics helps extract useful information from data. The Titanic dataset is widely used for understanding analytics and machine learning concepts.

ABOUT CODEALPHA

CodeAlpha provides internship opportunities focused on practical learning through real-world projects in Data Analytics and other technical domains.

PROBLEM STATEMENT

Analyze passenger information from the Titanic dataset and identify factors influencing survival using EDA and visualization.

OBJECTIVES

Understand the dataset, perform preprocessing, identify patterns, create visualizations, and draw conclusions.

DATASET DESCRIPTION

The Titanic dataset contains 891 records with passenger demographics, ticket details, and survival status.

TOOLS AND TECHNOLOGIES USED

Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab, and GitHub.

METHODOLOGY

Dataset Collection → Data Loading → Data Cleaning → EDA → Visualization → Interpretation → Conclusion.

DATA PREPROCESSING

The dataset was examined for missing values, checked for data types, and prepared for analysis.

EXPLORATORY DATA ANALYSIS

Analysis included survival distribution, gender-wise survival, passenger class analysis, and age distribution.

DATA VISUALIZATION

Created Survival Count Plot, Gender Survival Plot, Passenger Class Plot, Age Distribution Histogram, and Correlation Heatmap.

RESULTS AND FINDINGS

Female passengers showed higher survival rates. First-class passengers had better survival chances. Fare and class influenced outcomes.

CHALLENGES FACED

Handling missing values, selecting visualizations, and interpreting relationships between variables.

SKILLS ACQUIRED

Python Programming, Data Cleaning, EDA, Data Visualization, Analytical Thinking, and Report Writing.

REAL WORLD APPLICATIONS

Healthcare, Banking, E-commerce, and Transportation sectors use similar analytical techniques.

FUTURE SCOPE

Machine Learning prediction models, dashboards, advanced analytics, and real-time data applications.

CONCLUSION

The project successfully demonstrated EDA and Data Visualization techniques and identified key factors affecting survival.

REFERENCES

Kaggle Titanic Dataset, Python Documentation, Pandas Documentation, Matplotlib Documentation, Seaborn Documentation, Google Colab Documentation.